RIG-ETALE DESCENT ALONG ADIC MONOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: same shape as lemma-rig-etale-permanence (L48235); the adic-monomorphism hypothesis on g makes the square cartesian in the category of formal algebraic spaces

Editorial note. The label lemma-rig-etale-alternative-permanence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-base-change-rig-etale (delivered, L48219): rig-etale morphisms of locally Noetherian formal algebraic spaces are stable under base change

Source reference. Original reference 0GD1: sources/stacks-restricted/pinned-original-source.tex, source lines 5125–5125. Prerequisite pointer invoked at 5175; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of locally Noetherian formal algebraic spaces over S. If $g \circ f$ is rig-étale and g is an adic monomorphism, then f is rig-étale.*

Proof. Use Lemma 0GD1 and that f is the base change of $g \circ f$ by g. □

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.